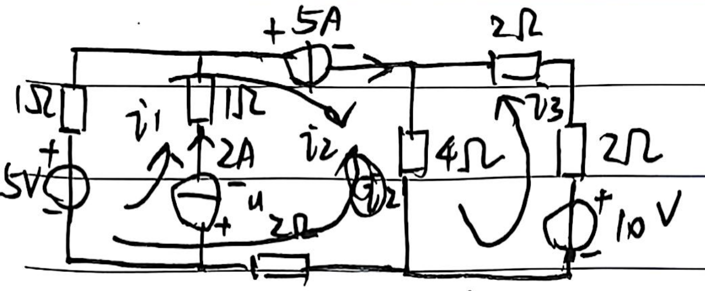
17



$i_1 = 2A$

$i_2 = 5A$

$\cancel{i_3 \cdot 8\Omega - 10V}$

$i_3 \cdot 8\Omega + i_2 \cdot 4\Omega - 10V = 0$

$\Rightarrow i_3 = 1.25A$

$\therefore P_{5V} = \cancel{i \cdot 5V}\ (i_1 - i_2) \cdot 5V = -15W$

$P_{10V} = -i_3 \cdot 10V = 12.5W$

$5 + u_{2A} + 2i_1 - i_2 = 0 \Rightarrow u_{2A} = -4V$

$\therefore P_{2A} = u_{2A} i_1 = -8W$

$u_{5A} + u_{2A} + 4i_2 + i_1 + 4i_3 = 0 \Rightarrow u_{5A} = -23V$

$\therefore P_{5A} = u_{5A} \cdot i_2 = -115W$

$u_a\ u_b\ u_c\ u_1\ i$

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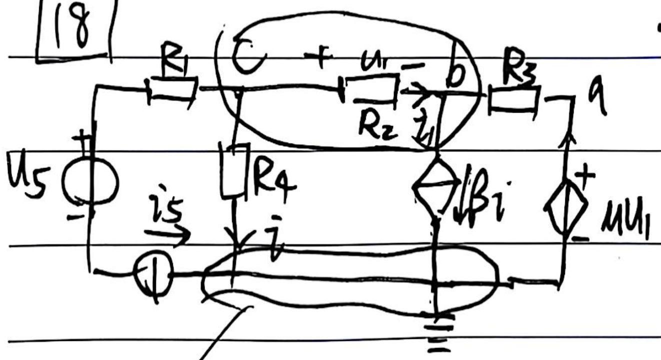

a: $u_a = \mu u_1$

b+c: $\frac{u_b - \mu u_1}{R_3} + \frac{u_c}{R_4} = -\beta i - i_S$

附: $u_c = i R_4$

$u_c - u_b = u_1$

u_c

$u_1 = i_1 R_2$

$i_1 = \beta i - \frac{\mu u_1}{R_3}$

$\Rightarrow u_1 = \beta i R_2 - \mu \frac{R_2}{R_3} u_1$ ①

KCL: $i + i_S + \beta i - \frac{\mu u_1}{R_3} = 0$ ②

$\Rightarrow u_1 = \frac{\cancel{R_2 R_3 i_S \beta}}{\cancel{R_3 \beta - R_3 - \mu R_2}}$ $\frac{\frac{\beta R_2}{1+\beta} i_S}{\frac{\beta R_2}{1+\beta}\frac{\mu}{R_3} - 1 - \frac{\mu R_2}{R_3}}$

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